Advanced approximate and symbolic computing

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Orthogonal transformations III.

application of QR decomposition, eigenvalue problem

Further notes on the QR decomposition

QR decomposition is not unique

Hint: consider how we obtain the element a_{11}.

Since the first column of matrix $\mathbf{R}$ is all zero except its first element, we have

$$a_{11} = q_{11}r_{11}.$$

Now, there are two possibilities:

- if $a_{11} > 0$ then either both q_{11} and r_{11} are positive or both are negative;
- if $a_{11} < 0$ then the signs of q_{11} and r_{11} need to be different.

QR factorization for non-square matrix

As we have already mentioned, the **QR** factorization can be done for *any* matrix.

For us, the important case is when matrix **A** has more rows than columns. The following theorem is regarding this case.

Theorem 5.1 *For any $m \times n$-dimensional matrix $\mathbf{A}$, where $m \geq n$, the factorization can be done in the form of*

$$\mathbf{A} = \mathbf{Q} \left(\frac{\mathbf{R}}{\mathbf{O}} \right) \begin{array}{r} n \ \text{ rows} \\ m-n \ \text{ rows} \end{array}$$

where $\mathbf{Q}$ is an $m \times m$-dimensional orthogonal matrix, and $\mathbf{R}$ is an $n \times n$-dimensional upper triangular matrix.

Applications of QR factorization

- (Computing the matrix determinant)
- (Inverting the matrix)
- Solving system of linear equations
  - there is a special case to be discussed here
- Most importantly: eigenvalues

Solving system of linear equations

- $\mathbf{A} = \mathbf{QR}$
- $\mathbf{Ax} = \mathbf{b} \Longleftrightarrow \mathbf{Q}(\mathbf{Rx}) = \mathbf{b}$, and use a new variable: $\mathbf{y} = \mathbf{Rx}$
- The decomposition can be applied as:

$$\mathbf{Ax} = \mathbf{b} \qquad \Longleftrightarrow \qquad \begin{cases} (i) & \mathbf{Qy} = \mathbf{b} \\ (ii) & \mathbf{Rx} = \mathbf{y} \end{cases}$$

- We can obtain $\mathbf{y} = \mathbf{Q}^T\mathbf{b}$ from (i) using $O(n^2)$ time.
- Then we obtain $\mathbf{x}$ from (ii) using backward substitution using $O(n^2)$ time.

Solving system of linear equations

- This method is particularly useful in case we have several equations with the same matrix.

- Given

$$
\begin{aligned}
\mathbf{A}\mathbf{x} &= \mathbf{b}_1 \\
\mathbf{A}\mathbf{x} &= \mathbf{b}_2 \\
&\vdots \\
\mathbf{A}\mathbf{x} &= \mathbf{b}_k
\end{aligned}
$$

it is more beneficial to first calculate the QR decomposition of matrix $\mathbf{A}$ and then solve the systems one-by-one:

$$
\begin{aligned}
\mathbf{Q}\mathbf{y} &= \mathbf{b}_j \\
\mathbf{R}\mathbf{x} &= \mathbf{y},
\end{aligned}
$$

where $j = 1, \ldots, k$.

- Consider: we need to calcuate the QR decomposition of $\mathbf{A}$ only once, and afterwards it is enough to have $\mathcal{O}(n^2)$ time.

- naively: $k \cdot \mathcal{O}(n^3)$
- cleverly: $\mathcal{O}(n^3) + k \cdot \mathcal{O}(n^2)$

Eigenvalues, eigenvectors

In the followings (unless we say something else) we mean

- matrix as in $\mathcal{C}^{n\times n}$(i.e. over complex numbers)
- vectors as in $\mathcal{C}^{n}$(i.e. vectors with complex numbers)

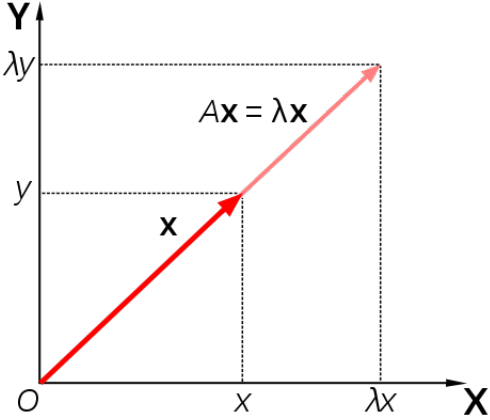
Let us start with the formal definition.

Definition 5.2 *Let* $\mathbf{A}$ *be a matrix. The number* $\lambda \in \mathcal{C}$ *is called the* eigenvalue *of matrix* $\mathbf{A}$*, if there is such vector* $\mathbf{v} \in \mathcal{C}^n$, $\mathbf{v} \neq \mathbf{0}$ *for which*

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v}$$

holds. Then we call vector $\mathbf{v}$ *as the* eigenvector *(corresponding to* λ*).*

Eigenvalue illustration



Animation: `http://upload.wikimedia.org/wikipedia/commons/a/ad/Eigenvectors-extended.gif`

Source: wikipedia

Eigenvalues, eigenvectors

- Since the eigenvalue problem leads to a problem of finding the roots of a polynomial of degree n, we know that an n-dimensional matrix always has n eigenvalues.

- For some matrix of special structure finding the eigenvalues are easy.

Proposition 5.3 *If* $\mathbf{R} \in \mathcal{C}^{n \times n}$ *is upper or lower triangular, then its eigenvalues are*

$$\lambda_k = r_{kk} \quad 1 \leq k \leq n.$$

This is very good!

Spectral radius and $\|.\|_2$ of matrices

- The spectral radius of matrix $\mathbf{A}$ is $\rho(\mathbf{A}) \stackrel{def}{=} \max_{1 \le i \le n} |\lambda_i|$,
- and the *trace* is $\operatorname{tr}(\mathbf{A}) \stackrel{def}{=} \sum_{j=1}^{n} a_{jj}$.

Now, we are ready to define the second norm of a matrix:

$$\|\mathbf{A}\|_2 \stackrel{def}{=} \sqrt{\rho(\mathbf{A}^T\mathbf{A})}$$

We can see that it is quite expensive to calculate the second norm.

The spectral radius $\rho(\mathbf{A})$ can be bounded by using any norm, as it is formalized in the following.

Proposition 5.4 *For any matrix* $\mathbf{A}$ *and any norm* $\|.\|$ *it inequality*

$$\rho(\mathbf{A}) \leq \|\mathbf{A}\| \tag{1}$$

holds.

Similarity transformations

One of the key definition in eigenvalue problems is similarity.

Definition 5.5 *Let* $\mathbf{A}$ *be a matrix and* $\mathbf{T}$ *be a regular matrix. The transformation*

$$\mathbf{A} \mapsto \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$$

is called similarity transformation.

We say that matrix $\mathbf{A}$ *is* similar *to matrix* $\mathbf{B}$, *if there exists a regular matrix* $\mathbf{T}$, *such that* $\mathbf{B} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}$.

- orthogonal similarity

- Given a vector space, if we change the basis vectors, then the matrix representation of a linear transformation also changes. If matrix $\mathbf{A}$ is corresponding to a linear transformation in a given basis, then the set

$$\{\mathbf{X}^{-1}\mathbf{A}\mathbf{X} \mid \mathbf{X} \in \mathcal{R}^{n\times n} \text{ is regular }\}$$

  contains all matrices which represent this linear transformation.

- This set is the set of matrices which are all similar to matrix $\mathbf{A}$.
- Similar matrices represent the same transformation in different basis.
- Hence, every linear transformation is equivalent to a class of matrices, so similarity is an equivalence-relation.

This will be denoted by: $\mathbf{A} \sim \mathbf{B}$.

- Some properties of linear transformations do not depend on the basis. Such properties are called *invariant* under similarity transformation.

- The most important fact with respect to eigenvalue problems is that the eigenvalues are invariant under similarity transformations.

Corollary 5.6 *Let* $\mathbf{A}, \mathbf{B} \in \mathcal{C}^{n\times n}$ *be arbitrary matrices.*

(i) *If* $\mathbf{A} \sim \mathbf{B}$*, then* $\lambda_j(\mathbf{A}) = \lambda_j(\mathbf{B})$*, i.e., their eigenvalues are equal.*

(ii) *If* $\mathbf{A} \sim \mathbf{B}$ *and matrix* $\mathbf{A}$ *has eigenvectors* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$*, then the eigenvectors of matrix* $\mathbf{B}$ *can be expressed in the form* $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$*, where*

$$\mathbf{v}_j = \mathbf{T}^{-1}\mathbf{u}_j$$

for any $1 \leq j \leq n$*.*

If the eigenvalues of two matrices **A** and **B** are the same, then this does not imply that they are similar.

Example 5.7 *It can be proved that matrix*

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

is not *similar to the identity matrix* **I**.

Triangularization

We say that matrix $\mathbf{A}$ can be *triangularized*, if it is similar to an upper-triangular matrix $\mathbf{R}$.

Theorem 5.8 *(Schur normal form of matrices) For any matrix* $\mathbf{A} \in \mathcal{C}^{n \times n}$ *there exists such unitary matrix* $\mathbf{U} \in \mathcal{C}^{n \times n}$ *where*

$$\mathbf{U}^H \mathbf{A} \mathbf{U} = \mathbf{R} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$$

holds. Meaning that matrix $\mathbf{A}$ *can be triangularized by unitary similarity transformations. The resulting triangular matrix* $\mathbf{R}$ *contains the eigenvalues of matrix* $\mathbf{A}$ *in its diagonal.*

What is a unitary matrix? $.^H$ is the conjugate transpose. They are the 'complex orthogonal' matrices

So far. . .

- . . . we have seen that an algorithm *could* be that we choose appropriate matrices in order to do similarity transformations which makes matrix $\mathbf{A}$ into triangular form.
- It is important to note here that we could only state existence theorems.
- Is there a method which can produce these matrices in finite steps?
- We have already mentioned the theorem of Abel-Ruffini, which implies that we have no hope.
- Anyway, let us see an efficient approximation scheme.
- That is going to be explained in the next lecture.